

BALSAM: A Platform for Benchmarking Arabic Large Language Models

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Abstract

The impressive advancement of Large Language Models (LLMs) in English has not been matched across all languages. In particular, LLM performance in Arabic lags behind, due to data scarcity, linguistic diversity of Arabic and its dialects, morphological complexity, etc. Progress is further hindered by the quality of Arabic benchmarks, which typically rely on static, publicly available data, lack comprehensive task coverage, or do not provide dedicated platforms with blind test sets. This makes it challenging to measure actual progress and to mitigate data contamination. Here, we aim to bridge these gaps. In particular, we introduce BALSAM, a comprehensive, community-driven benchmark aimed at advancing Arabic LLM development and evaluation. It includes 78 NLP tasks from 14 broad categories, with 52K examples divided into 37K test and 15K development, and a centralized, transparent platform for blind evaluation. We envision BALSAM as a unifying platform that sets standards and promotes collaborative research to advance Arabic LLM capabilities.¹

1 Introduction

Arabic is a prominent language with more than 400 million speakers (Boulesnam and Boucetti, 2025) and major religious significance for two billion Muslims. This has translated into significant demand for robust Arabic Natural Language Processing (NLP) systems, resulting in the development

of multiple Arabic-centric Large Language Models (LLMs), such as Jais (Sengupta et al., 2023) and Fanar (Fanar Team et al., 2025), and in improved Arabic support in multilingual models such as Gemini (Gemini Team et al., 2023), GPT-4o (OpenAI et al., 2024). Despite recent progress, LLMs still underperform in Arabic compared to English. This stems from limited training data, the linguistic diversity of Modern Standard Arabic (MSA) and regional dialects, and Arabic’s complex morphology.

Robust benchmarking is crucial to quantify the gaps and guide future improvements in Arabic capabilities of LLMs. Yet, existing Arabic benchmarking initiatives, such as LARA-Bench (Abdelali et al., 2024), have primarily focused on standard natural language generation and understanding tasks. A more recent effort, AraGen (El Filali et al., 2024), introduced a leaderboard-based framework that evaluates LLM performance across multiple dimensions, including correctness, completeness, conciseness, helpfulness, honesty, and harmlessness, in an LLM-as-a-judge setup. In parallel, several datasets have been developed to assess LLM capabilities across different dimensions: ArabicMMLU (Koto et al., 2024) targets world knowledge, AraDICE (Mousi et al., 2025) focuses on dialects with cognitive and cultural understanding, Palm (Alwajih et al., 2025) addresses cultural comprehension, and Ashraf et al. (2025) focus on safety. However, existing efforts address limited LLM capabilities, lack comprehensive coverage, and have no dedicated *platforms* for community

¹<https://benchmarks.ksaa.gov.sa>

collaboration. Critically, measuring progress in a consistent and reliable manner requires a standardized, community-driven framework with blind test datasets, an aspect that remains largely lacking.

Here, we aim to bridge this gap. In particular, we present the *Benchmark for Arabic Language Models (BALSAM)*, which is a comprehensive community-driven initiative designed to advance benchmarking efforts for Arabic LLMs. *BALSAM* includes a collection of 78 tasks across 14 categories, with a total of 52K examples divided into 37K test and 15K dev. These tasks span a wide range of natural language understanding and generation tasks, including summarization, question answering, information extraction, machine translation, and text classification, among others.

BALSAM further provides an integrated *evaluation platform* featuring an Arabic LLM Leaderboard. This enables the research community to systematically assess the performance of Arabic LLMs, to monitor progress over time, and to access up-to-date benchmark results for the top-performing LLMs. The *BALSAM* platform goes beyond a traditional leaderboard, serving as a collaborative effort for leading academic and governmental institutions across the Middle East and beyond. Its core mission is to drive the creation of domain-specific test datasets and to establish robust benchmarks for evaluating Arabic LLMs. By promoting transparency and cooperation, *BALSAM* aims to unify the Arabic NLP community around shared datasets and standards. Further, we investigate a variety of automated metrics and measure their correlation with human evaluation. We show that using LLM-as-a-Judge highly correlates with human judgments while other measures such as BLEU, ROUGE, and BertScore don't.

The contributions of *BALSAM* and this paper are summarized as follows:

- *BALSAM* is a community driven consortium that provides a centralized evaluation platform with an associated leaderboard.
- *BALSAM* provide diverse dev/test sets based on 78 tasks, where the test sets are blind.
- We compare the efficacy of using automated evaluations based on BLEU, ROUGE, BERTScore, and LLM-as-a-judge compared to human judgments.

2 Related Work

This section reviews prior work across four dimensions: Arabic-centric benchmarks developed to evaluate LLMs in MSA and dialects, English and multilingual benchmarks providing broader frameworks but with limited Arabic coverage, tools and leaderboards enabling systematic model comparison, and a concluding Challenges and Gaps subsection that synthesizes the main limitations of earlier efforts

2.1 Arabic-Centric Benchmarks

Recent efforts have focused on benchmarking LLMs for Arabic, targeting tasks such as natural language understanding, generation, and speech processing (Abdelali et al., 2024; Elmadany et al., 2023; Nagoudi et al., 2023). While LLMs have demonstrated remarkable capabilities across various domains, including solving graduate-level mathematical problems and passing medical examinations, these achievements have been predominantly assessed using English-language benchmarks. Thus, in order to evaluate and advance the performance of LLMs for Arabic, there is a critical need for the development of dedicated Arabic benchmarks. Koto et al. (2024) developed ArabicMMLU, an Arabic version of the MMLU benchmark constructed from authentic school exam questions sourced from Arabic-speaking countries, without relying on translation. Similarly, Mousi et al. (2025) created resources for MSA and dialectal Arabic, aiming to assess linguistic, cognitive, and cultural competencies. Alwajih et al. (2025) introduced datasets to evaluate the cultural and dialectal capabilities of LLMs. Almazrouei et al. (2023) adopted and restructured existing datasets to create benchmarks for evaluating LLMs in MSA and dialectal Arabic. Moreover, resources have been developed to assess domain-specific knowledge, e.g., ArabLegalEval (Hijazi et al., 2024) focuses on legal knowledge, while Qiyas (Al-Khalifa and Al-Khalifa, 2024) targets mathematical reasoning. Finally, Ashraf et al. (2025) developed an Arabic dataset for safety.

2.2 English/Multilingual Benchmarks

Several prominent benchmarks remain focused on English-centric evaluations, including MMLU (Hendrycks et al., 2021), HELM (Liang et al., 2023), and BIG-bench (Srivastava et al., 2022). MMLU is designed to assess reasoning and knowl-

edge in real-world contexts, while HELM evaluates LLMs across a variety of metrics and scenarios. BIG-bench offers an extensive evaluation framework comprising 214 tasks, some of which include coverage of low-resource languages. Additionally, a range of multilingual benchmarks have been developed to assess model performance across diverse languages, including morphologically complex and low-resource languages such as Arabic.

2.3 Tools and Leaderboards

As LLMs continue to advance rapidly, it has become essential to compare their performance across various capabilities and domains. Over time, numerous tools and leaderboards have been developed to facilitate such evaluations. This includes [LLMeBench](#), a comprehensive benchmarking platform with a primary focus on Arabic NLP, speech, and multimodal tasks (Dalvi et al., 2024). Moreover, tools such as [LM-Evaluation-Harness](#), [OpenCompass](#), and [BigCode-Evaluation-Harness](#) provide standardized frameworks for assessing model performance across a wide range of tasks and datasets, facilitating more robust and comprehensive comparisons, as well as signaling to LLM developers areas in which their models need improvement. Several open-source leaderboard initiatives have emerged to benchmark Arabic language models, including the [Open Arabic LLM Leaderboard](#), the [Arabic-MMMLU-Leaderboard](#) (Nacar et al., 2025), and [AraGen](#) (El Filali et al., 2024). Each of them serves a specific purpose. For example, the Arabic-MMMLU-Leaderboard is based on the MMMLU OpenAI benchmark, while AraGen focuses on a diverse set of tasks such as question answering, summarization, and reasoning.

2.4 Challenges and Gaps

Existing evaluation benchmarks rely on static, publicly available datasets, enabling rapid community assessment. Yet, as LLMs advance rapidly, static benchmarks struggle to capture their evolving capabilities. The growing size of LLMs and their increasingly extensive training data heighten the risk of test data contamination, which is difficult to detect due to opaque training data and widespread use of synthetic data (Dong et al., 2024). Hence, leaderboards with rigorous contamination checks and adaptive benchmarks that reflect the latest model capabilities are needed (Deng et al., 2023).

The LMSYS Chatbot Arena (Zheng et al., 2023; Chiang et al., 2024) enables robust evaluation of

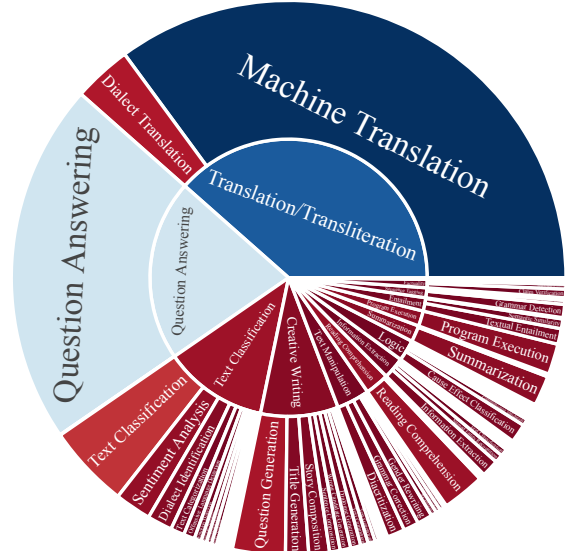


Figure 1: *BALSAM* data distribution across general categories and tasks in these categories.

LLMs through conversational interactions and Elo-based rankings (Bai et al., 2022), but relies on human evaluation, which is time-consuming and limits scalability (Luo et al., 2024). The LLM-as-a-judge approach was introduced to reduce human involvement on platforms such as Chatbot Arena and MT-bench (Zheng et al., 2024), but it requires careful handling to avoid biases such as verbosity, position, and self-enhancement. Moreover, this method struggles with assessing reasoning and math tasks. Indeed, several popular leaderboards, including MT-bench and OpenLLM, face issues of saturation and inconsistent alignment with real-world chatbot performance (Luo et al., 2024).

Despite significant progress in developing English benchmarks and LLM leaderboards, there remains much work to be done for languages such as Arabic. This includes the creation of new datasets to address emerging capabilities and the establishment of sustainable leaderboards that integrate human and LLM-based evaluation approaches.

3 BALSAM Dataset

3.1 Dataset Creation

The *BALSAM* benchmark is composed of 78 tasks from 14 coarse-grained categories, with a total of 52K examples divided 37K test and 15K development, and a centralized, transparent platform for blind evaluation. We made the design decision to have many datasets, but only have 10–100 test examples per dataset. For most datasets, we also have up to 50 development examples.

Figure 1 shows the data distribution across general categories and tasks in these categories. We can see that the main categories are multiple-choice questions (MCQ), text generation, translation, and transliteration. Table 7 and Table 8 in the Appendix gives the complete list of tasks in *BALSAM* along with the sizes of their development and test sets. The number of examples varies widely between tasks, with some tasks containing thousands of samples and others only a few. Figure 2 in the Appendix shows sample entries for different categories. Note that we converted some tasks to MCQ or text generation, e.g. Part-of-Speech (POS) tagging and Named Entity Recognition (NER), which have been traditionally addressed as sequence labeling tasks. The aim was to ease evaluation as we currently cannot handle sequence labeling tasks (we plan support for this in the future).

Reusing Public Datasets Some of the datasets are subsampled from publicly available test sets with preexisting prompts and ground-truth answers. This includes datasets from the Arabic subset of the xP3 dataset (Muennighoff et al., 2023), from which we subsampled 68 datasets, covering 12 tasks, to include 25 development and 50 test examples. We further reformatted AraMath (Alghamdi et al., 2022) to MCQ format, as an additional dataset.

Prompting Existing NLP Datasets We created natural language prompts based on publicly available Arabic NLP datasets using the PromptSource tool (Bach et al., 2022). We developed 2–8 different prompt templates per dataset, resulting in an equal number of sub-datasets. Figure 3 in the Appendix shows four different prompt templates developed for one of the datasets.

Translating English Datasets to Arabic Some of our datasets were created by translating existing English datasets to Arabic. We have a total of 483 such datasets, covering 29 different tasks, sampled from PromptSource (Bach et al., 2022), Super-NaturalInstructions (Wang et al., 2022; Mishra et al., 2022), and TruthfulQA (Lin et al., 2022). The translations were evaluated both automatically and manually as described in (El-Sheikh et al., 2024).

Developing New Datasets We further developed 16 brand new datasets with 1,755 prompts, covering specialized, structured, and rare examples to better test model generalization, e.g., to tasks such as grammatical error detection and factuality.

Augmenting with Synthetic Examples Our target was to have 10–100 test examples per dataset. However, for 14 datasets, we had less than 10 examples; we thus used GPT-4o to generate synthetic examples, which we checked manually.

3.2 Quality Assurance

To ensure data quality, we conducted extensive quality checks in three iteratively repeated stages:

- *Completeness:* We ensured that all required fields in all datasets were fully populated, with no missing or null values. We found that 1% of our test examples contained null values, which we removed; we further found that 7% of the datasets included duplicates, which we also removed.
- *Consistency:* We established a standardized format to maintain consistency across the datasets. We found that approximately 17% of the datasets exhibited format-related issues, such as improper structure, or incorrect labels, which we fixed.
- *Reliability* We asked 16 annotators to conduct a manual review of random samples from each dataset checking that each *instruction*, *input*, and *output* were clear and cultural appropriate. We found issues for 10% of the datasets; to fix them, we edited some specific examples or excluded entire datasets.

3.3 Mitigating Data Leakage

A primary goal of the *BALSAM* initiative is to establish a fair, unbiased, and trusted benchmark for evaluating LLMs in Arabic. Thus, it is critical to prevent test set leakage and to minimize the risk of contamination of LLM training data.

In order to protect the integrity and reliability of the benchmark, we restricted the access to the test sets to a small group of individuals responsible for quality assessment and platform development: in fact, the vast majority of members of the *BALSAM* team only know the part of the raw test data candidates they contributed initially, but they have no access to the final test data.

4 Evaluation Setup

4.1 Benchmarking Phases

The *BALSAM* benchmark comprises a total of 37,419 test and 15,742 development examples and runs in two phases:

- *Phase 1.* This phase includes 54 tasks across 13 categories focusing on text generation. It contains 13,121 test and 6,434 dev examples. The largest categories are *creative writing* and *translation*, which cover tasks such as *story composition* and *dialect translation*, respectively. A complete breakdown of the categories and associated tasks in this phase is given in Table 7 in the Appendix.
- *Phase 2:* This phase includes 50 tasks across 13 categories and contains 24,298 test examples and 9,308 development examples. The focus of this phase is on multiple-choice question answering and specific generation tasks (Diacritization, Translation/Transliteration).

The two phases share 12 categories in common, with the remaining categories being *translation* (unique to Phase 1) and *factuality* (unique to Phase 2). A complete breakdown of all categories and tasks is provided in Table 8 in the Appendix.

4.2 Evaluation Framework

We adopted the LM-Evaluation-Harness (Gao et al., 2024) framework, henceforth *LM-Harness*, for several reasons: (i) it supports evaluation of both open-source LLMs with accessible weights as well as commercial LLMs that are only available via API calls, (ii) it allows flexible customization of tasks and benchmarks through YAML files, and (iii) it has been used in various leaderboards on Hugging Face and as part of various LLM development pipelines, e.g., by Fanar (Fanar Team et al., 2025).

4.3 Evaluation Platform

We enhanced the schema of *LM-Harness*² to standardize the input data. Each dataset file is assigned a unique ID, and its JSON content is preprocessed into the YAML format required by *LM-Harness*, which includes task metadata and dataset split paths. The evaluation jobs on the platform are organized into categories, tasks, and datasets. Categories group related tasks for visualization purposes. Tasks represent specific objectives such as summarization, sequence tagging, title generation, and transliteration, while datasets contain data split by prompts and data items for each task.

Users register models via an OpenAI-compatible API (requiring model ID and URL) or a public model (e.g., from aiXplain) with optional metadata

such as model name and training data. Evaluation requests are run in parallel for selected categories to minimize waiting times. Results are calculated as task-level macro-averages of dataset scores. Similarly, category-level results are computed as the macro-average of per-task scores. The overall score of a model is the macro-average score across all tasks. The *BALSAM* Leaderboard³ summarizes the model performance, displaying average scores for all tasks. Scores, ranging from 0 to 1, reflect task-specific metrics and enable clear comparisons of model performance across tasks.

4.4 Evaluation Measures

Given that the focus of Phase 1 on text generation, we began evaluation using BLEU (Papineni et al., 2002) for the translation category and ROUGE-LSum for the rest of categories (Lin, 2004). For analysis purposes, we also perform manual judgments (see below).

5 Experiments

5.1 Experimental Setup

We selected a comprehensive set of LLMs that support Arabic; see Appendix D for a detailed list and description of the models we used.

- *Open-weights models:* we chose them based on public availability, relevance to Arabic NLP, and architectural diversity. We conducted all experiments using four NVIDIA A100 GPUs, each with 40G of VRAM.
- *Closed models:* we included some popular ones that support Arabic, and we accessed them via their standard APIs or by provider request.

5.2 Results and Discussion

Challenges in Automatic Evaluation. Table 1 shows the automatic evaluation results of the LLMs across 13 categories using ROUGE-LSum and BLEU. Unexpectedly, the results show that SILMA-9B is far ahead of much larger models such as Aya 32B, Qwen-2.5 32B, and DeepSeek V3. This prompted us to manually examine random output samples to better understand the underlying reasons. Our analysis revealed the following:

²<https://github.com/ksaa-nlp/balsam-eval>

³<https://benchmarks.ksaa.gov.sa/b/balsam>

Model	CW	ENT	FIB	IE	LOG	PE	QA	RC	ST	SUM	TC	TM	MT/TL	AVG	AVG*
SILMA-9B Instruct-v1.0	0.23	0.13	0.12	0.32	0.22	0.66	0.31	0.55	0.20	0.20	0.36	0.60	0.13	0.31	0.33
Nuha v2	0.22	0.12	0.12	0.32	0.20	0.81	0.25	0.35	0.28	0.19	0.39	0.64	0.15	0.31	0.32
Jais-family 13B-chat	0.24	0.08	0.10	0.25	0.17	0.89	0.22	0.51	0.19	0.26	0.15	0.48	—	—	0.30
Command R+	0.19	0.07	0.11	0.33	0.17	0.68	0.31	0.41	0.28	0.16	0.28	0.53	0.15	0.28	0.29
GPT-4o	0.22	0.10	0.20	0.28	0.16	0.21	0.23	0.29	0.38	0.17	0.30	0.62	0.17	0.26	0.27
Iron Horse GV V5a	0.20	0.10	0.21	0.27	0.15	0.48	0.21	0.24	0.36	0.15	0.27	0.56	0.14	0.26	0.27
Yehia 7B Preview	0.23	0.13	0.18	0.26	0.20	0.34	0.23	0.28	0.26	0.20	0.24	0.62	0.14	0.25	0.27
AceGPT-v2 8B Chat	0.19	0.11	0.14	0.29	0.18	0.49	0.25	0.36	0.19	0.19	0.23	0.51	0.11	0.25	0.26
Grok-2-latest	0.20	0.08	0.14	0.23	0.15	0.16	0.22	0.29	0.30	0.18	0.18	0.49	0.14	0.21	0.24
Gemini 2.0 Flash	0.17	0.06	0.14	0.28	0.15	0.13	0.25	0.30	0.33	0.15	0.24	0.33	0.13	0.20	0.22
Mistral-saba-latest	0.21	0.07	0.16	0.18	0.14	0.15	0.20	0.23	0.29	0.18	0.19	0.55	0.15	0.21	0.21
Claude Sonnet 3.5	0.13	0.15	0.07	0.19	0.09	0.24	0.18	0.20	0.35	0.15	0.12	0.38	0.12	0.18	0.19
Command-r7b 12-2024	0.17	0.07	0.15	0.15	0.13	0.26	0.15	0.22	0.19	0.16	0.13	0.41	0.13	0.18	0.19
Gemma 2 9B	0.16	0.09	0.11	0.19	0.14	0.31	0.18	0.23	0.19	0.15	0.10	0.30	0.05	0.17	0.19
Qwen 2.5 32B	0.14	0.09	0.13	0.16	0.11	0.30	0.15	0.13	0.23	0.16	0.08	0.43	0.08	0.17	0.18
DeepSeek V3	0.17	0.12	0.11	0.18	0.11	0.12	0.15	0.14	0.25	0.15	0.08	0.40	0.15	0.16	0.17
C4AI Aya Expanse 32B	0.14	0.07	0.11	0.13	0.07	0.23	0.14	0.25	0.13	0.19	0.06	0.38	0.10	0.15	0.16
Fanar-C-1-8.7B	0.14	0.09	0.07	0.16	0.11	0.36	0.14	0.15	0.11	0.14	0.11	0.33	—	—	0.16
Amazon Nova Pro	0.15	0.07	0.07	0.12	0.07	0.14	0.15	0.10	0.26	0.15	0.04	0.37	0.09	0.14	0.15
Mistral Large	0.08	0.10	0.04	0.10	0.08	0.17	0.12	0.15	0.06	0.07	0.09	0.30	0.05	0.11	0.12
DBRX-instruct	0.03	0.01	0.03	0.03	0.02	0.10	0.04	0.04	0.03	0.03	0.02	0.12	0.02	0.04	0.04
Aragpt2 mega	—	0.11	0.04	—	0.04	—	0.05	0.06	0.04	0.13	0.06	0.33	—	—	—

Table 1: **Automatic evaluation across categories.** “—” indicates that the model exceeded the token limits and did not complete the category. List of categories: CW (Creative Writing), ENT (Entailment), FIB (Fill in the Blank), IE (Information Extraction), LOG (Logic), PE (Program Execution), QA (Question Answering), RC (Reading Comprehension), ST (Sequence Tagging), SUM (Summarization), TC (Text Classification), TM (Text Manipulation), MT/TL (Machine Translation/Transliteration), AVG (Average), AVG* (Average w/o Translation).

- SILMA-9B’s output was generally terse, while the outputs of the other models were verbose; the metrics naturally preferred shorter answers. In a Question Answering example where the correct answer was باريس (Paris), SILMA-9B gave a matching terse reply, while other models provided more detailed, verbose answers with 25 words or longer (Full example in Appendix E).
- BLEU uses the geometric mean of unigram to 4-gram precisions. Because many gold answers were short, trigram and 4-gram matches were often absent, causing BLEU scores to be zero despite matching unigrams and bigrams.
- BLEU and ROUGE rely on exact word matches, which is difficult for Arabic’s complex morphology. For example, the reference كتاب (‘book’) and the prediction الكتاب (‘the book’) do not match exactly.

Human Evaluation. Next, we conducted a manual evaluation on a random sample of the test set, composed of 20 questions per category, where humans would rate the outputs from all LLMs. The correctness of each output, on a 0–3 scale, was judged by three judges. Thus, the total number of performed judgments was $254 \text{ questions} \times 22 \text{ LLMs} \times 3 \text{ judges} = 16,764 \text{ judgments}$. The de-

tailed annotation instructions we gave to the judges are given in Appendix G.

The average score per model from these judgments are reported in Table 2, where we can see that GPT-4o achieves the highest average score.

Human-to-Automatic Measure Correlation.

We measured the Pearson correlation of human judgments against ROUGE-LSum and BLEU. Table 3 shows the correlation between the three human judgments across categories. The average correlation between the judges is 0.75, and they correlated more with each other for some categories compared to others. For example, Creative Writing had the lowest correlation (0.636), while Reading Comprehension had the highest correlation (0.88). Table 4 lists the correlations of manual evaluation against ROUGE-LSum and BLEU. Since we had three judges, we computed the correlation between the metrics and the average judges’ scores. We can see very poor correlation between manual judgments and automatic measures.

Beyond BLEU and ROUGE. We explored some alternative evaluation approaches, namely:

- **Semantic Evaluation:** We used BERTScore (Zhang et al., 2020), which captures semantic similarity more effectively than surface-level n -gram overlap.

Model	CW	ENT	FIB	IE	LOG	PE	QA	RC	ST	SUM	TC	TM	MT/TL	AVG	AVG*
GPT-4o	2.78	3.00	2.50	2.57	2.50	2.30	2.30	2.65	2.12	2.72	2.57	2.72	2.75	2.58	2.56
Iron Horse GV V5a	2.63	3.00	2.50	2.32	2.15	2.50	2.25	2.77	2.08	2.52	2.23	2.52	2.85	2.49	2.46
Claude Sonnet 3.50	2.83	2.52	2.57	2.58	2.27	2.58	2.35	2.53	2.02	2.62	1.97	2.67	2.68	2.48	2.46
DeepSeek V3	2.70	2.93	2.17	2.57	2.20	2.30	2.37	2.80	1.97	2.88	1.87	2.52	2.48	2.44	2.44
Nuha v2	2.75	2.62	2.53	2.38	1.95	2.20	2.32	2.83	2.02	2.88	1.70	2.70	2.63	2.42	2.40
Grok-2-latest	2.78	3.00	1.95	2.47	2.52	2.50	2.27	2.80	1.80	2.75	1.60	2.47	2.53	2.42	2.41
Gemini 2.0 Flash	2.73	2.74	2.42	2.53	2.43	2.13	2.12	2.77	1.95	2.60	1.55	2.37	2.72	2.39	2.36
Command R+	2.60	2.81	2.23	2.52	2.13	2.08	2.30	2.58	1.97	2.70	1.57	2.37	2.42	2.33	2.32
Fanar-C-1-8.7B	2.73	2.98	1.82	2.62	2.25	2.25	2.70	2.82	1.22	2.67	1.62	2.03	—	—	2.31
c4ai-aya-expense-32b	2.65	2.88	2.37	2.37	2.02	2.28	2.23	2.57	1.68	2.70	1.62	2.47	2.13	2.31	2.32
Mistral-saba-latest	2.60	2.86	2.15	2.55	2.00	1.25	2.38	2.82	1.90	2.78	1.43	2.53	2.50	2.29	2.27
Yehia-7B preview	2.68	2.98	1.88	2.28	2.08	1.83	2.28	2.63	1.75	2.50	1.65	2.68	2.13	2.26	2.27
Amazon Nova Pro	2.65	2.86	2.23	2.20	2.18	1.42	2.32	2.63	1.78	2.75	1.60	2.35	2.42	2.26	2.25
Gemma2 9B	2.62	2.90	1.70	2.33	2.08	1.97	2.20	2.85	1.73	2.67	1.77	1.93	2.05	2.22	2.23
Qwen-2.5 32b	2.83	2.55	1.97	2.15	1.97	2.18	2.12	2.72	1.77	2.55	1.45	2.42	2.08	2.21	2.22
Command-r7b 12-2024	2.62	2.83	1.60	2.08	1.88	2.00	2.20	2.45	1.75	2.77	1.18	2.38	1.87	2.12	2.15
Jais-family 13b-chat	2.03	2.88	1.13	2.23	1.70	2.17	1.87	2.52	1.35	2.38	1.02	2.18	—	—	1.96
SILMA-9B Instruct-v1.0	2.33	2.00	1.42	2.1	1.73	1.68	1.83	2.13	1.52	2.4	1.63	2.28	2.00	1.93	1.92
AceGPT-v2-8B-Chat	2.17	2.21	1.08	2.17	1.75	1.38	1.50	2.57	1.63	2.62	1.07	2.05	1.77	1.84	1.85
Mistral large	1.20	1.79	0.80	0.98	1.22	0.98	1.65	1.52	0.65	0.58	0.62	1.27	1.78	1.16	1.11
DBRX-instruct	0.23	0.24	0.07	0.22	0.28	0.73	0.43	0.18	0.77	0	0.22	0.12	1.28	0.37	0.29
Aragpt2-mega	—	0.14	0.13	—	0.13	—	0.13	0.42	0.1	1.63	0.05	0.37	—	—	—

Table 2: **Manual evaluation (3 evaluators; 20 examples per category).** “—” indicates that the model exceeded token limits and did not complete the category. List of categories: CW (Creative Writing), ENT (Entailment), FIB (Fill in the Blank), IE (Information Extraction), LOG (Logic), PE (Program Execution), QA (Question Answering), RC (Reading Comprehension), ST (Sequence Tagging), SUM (Summarization), TC (Text Classification), TM (Text Manipulation), MT/TL (Machine Translation/Transliteration), AVG (Average), AVG* (Average w/o Translation).

- **LLM-Based Answer Extraction:** We used Gemini 2.5 Flash (zero-shot, no chain-of-thought) to extract concise answers from the model-generated outputs. We used the prompt reported in the Appendix, Listing 1.
- **LLM-Based Scoring:** We used Gemini 2.5 Flash to rate the extracted answers on a 0–3 scale, mirroring the manual evaluation scheme.⁴ The scoring prompt is shown in Appendix Listing 2.

Table 5 shows the correlation of human evaluation with ROUGE-LSum, BLEU, and BERTScore (with and without extraction of answers using an LLM) and LLM as a judge. We make the following observations:

- Using an LLM to extract the answer from the LLM output generally had a positive impact on correlation for all measures (ROUGE-LSum, BLEU, and BERTScore).
- BERTScore correlated better with human judgments compared to ROUGE and BLEU.

⁴We also experimented with GPT-4o and GPT-4o mini as LLM judges. GPT-4 and Gemini showed nearly identical correlation with human scores, both outperforming GPT-4o mini by a sizable margin. Eventually, we selected Gemini 2.5 Flash due to its substantially lower cost.

Category	1 & 2	1 & 3	2 & 3	Avg.
Creative Writing	0.579	0.563	0.765	0.636
Entailment	0.824	0.757	0.768	0.783
Fill in the Blank	0.587	0.659	0.826	0.691
Info. Extraction	0.636	0.602	0.730	0.656
Logic	0.578	0.630	0.586	0.598
Program Execution	0.722	0.697	0.841	0.753
Q&A	0.883	0.813	0.816	0.837
Reading Compr.	0.885	0.894	0.860	0.880
Sequence Tagging	0.738	0.768	0.935	0.814
Summarization	0.828	0.774	0.754	0.785
Text Classification	0.833	0.820	0.921	0.858
Text Manipulation	0.790	0.808	0.792	0.797
Translation	0.646	0.607	0.746	0.666
Average	0.733	0.722	0.795	0.750

Table 3: Correlation between the three human judges (1, 2, & 3) per category.

- The correlation for ROUGE-LSum, BLEU, and BERTScore varied widely from category to category, and the average was low.
- LLM as a judge was highly correlated with human judgments for all categories, with values ranging between 0.824 and 0.977. In fact, it correlated better with the average of judges’ scores than judges correlated with each other.

Based on the above, we decided to drop ROUGE, BLEU, and BERTScore and rely solely on LLM as a Judge to evaluate the LLMs. Table 6 lists the results for all models on the entire *BALSAM* test set using LLM as a judge. When comparing the results

Category	ROUGE-Lsum	BLEU
Creative Writing	-0.509	-0.613
Entailment	-0.300	0.010
Fill in the Blank	-0.033	-0.008
Info. Extraction	0.139	0.514
Logic	0.425	0.296
Program Execution	-0.151	-0.005
Question Answering	0.339	0.316
Reading Comprehension	0.318	0.299
Sequence Tagging	0.537	0.094
Summarization	-0.393	-0.187
Text Classification	0.100	0.090
Text Manipulation	0.462	0.460
Translation	0.506	0.481
Average	0.111	0.134

Table 4: Correlation of human judgments against ROUGE-LSum and BLEU for different categories.

of using ROUGE-LSum and BLEU (Table 1) to using LLM as a judge (Table 6), we can see that the order of LLMs changes completely. In fact, the top performer in Table 1, namely SILMA-9B-IT came out in the lower third in Table 6. Given the aforementioned discussion, the LLM as a judge results are more trustworthy as they correlate much better with human judgments.

The results show that large closed models, e.g. GPT-4o, Gemini 2.0, and DeepSeek V3, significantly outperform all smaller Arabic-centric models such as Jais and Fanar. Two large models, namely Mistral large and DBRX-instruct performed poorly, trailing most others. Hence, model size is not a sufficient predictor of performance. Some of the most likely factors that come into play are Arabic tokenization, size of Arabic training set, and Arabic-centric supervised fine-tuning.

The results show some variability of how models generally perform for certain categories compared to others. For example, models overall perform better on some tasks, such as *translation* and *entailment*, and worse on others, such as *fill in the blank*. Some models are relatively more capable for some categories compared to others. For example, Grok-2 leads the pack for Logic and Iron Horse leads for Program Execution. Similarly, some models rank higher for some categories and much lower in others. For example, Jais and Fanar performed well for Summarization but poorly for Sequence Tagging. Some models performed poorly across the board, such as Aragt2-mega and DBRX-Instruct.

6 Conclusion and Future Directions

We have presented *BALSAM* — a major collaborative effort to establish benchmarking standards and foster unity in LLM development and evaluation for Arabic. *BALSAM* marks a significant step forward, offering evaluation across 78 tasks from 14 categories, with 37K development and 15K test examples. It further offers an integrated platform, and Arabic LLM Leaderboard that enable effective evaluation, comparison, and progress tracking with reliable LLM-as-a-judge based evaluation. However, challenges remain in enhancing data quality, addressing Arabic’s linguistic diversity, and expanding the scope of tasks covered.

In future work, we aim to improve dataset quality (e.g., eliminate translations and any form of synthetic data generation) to add additional tasks, as well as to address the limitations listed in the next section.

Limitations

Our study provides insights into LLM performance; however, several key limitations warrant consideration and will be the focus of the next iteration of the *BALSAM* benchmarking test sets.

- Token length restrictions in certain models precluded their complete participation across all evaluation tasks, particularly affecting models with restricted context windows and preventing calculation of comprehensive performance scores for these systems.
- While efforts have been made to ensure the accuracy and neutrality of the datasets, we acknowledge the potential for unintended biases, particularly those arising from translated datasets that may have translation errors or cultural misalignments. For example, certain phrases, such as “the Messenger of Islam Muhammad” were identified as potentially problematic, as they may not align with widely accepted terminologies within specific cultural and religious contexts, such as the more commonly used “Prophet Muhammad” in Arabic and Islamic discourse.
- Though *BALSAM* benchmarks LLMs across a variety of categories, some notable other functions and features of LLMs need to be considered such as fluency of the generated output, cultural alignment, ability to answer

Category	ROUGE	Ext. ROUGE	BLEU	Ext. BLEU	BERT	Ext. BERT	LLM-J
Creative Writing	-0.509	-0.476	-0.613	-0.582	-0.629	-0.392	0.824
Entailment	-0.300	0.227	0.010	0.546	-0.244	-0.176	0.950
Fill in the Blank	-0.033	0.390	-0.008	-0.502	0.386	0.696	0.944
Information Extraction	0.139	0.656	0.514	0.766	0.034	0.691	0.824
Logic	0.425	0.742	0.296	0.554	0.429	0.676	0.945
Program Execution	-0.151	0.715	-0.005	0.882	-0.235	-0.034	0.911
Question Answering	0.339	0.807	0.316	0.494	0.408	0.852	0.977
Reading Comprehension	0.318	0.285	0.299	0.008	0.413	0.268	0.931
Sequence Tagging	0.537	-0.241	0.094	-0.793	0.691	0.182	0.931
Summarization	-0.393	-0.754	-0.187	-0.676	0.092	-0.604	0.934
Text Classification	0.100	0.400	0.090	0.275	0.251	0.830	0.948
Text Manipulation	0.462	0.677	0.460	0.685	0.401	0.678	0.919
Translation	0.506	0.806	0.481	0.754	0.390	0.831	0.899
Average	0.111	0.326	0.134	0.186	0.184	0.346	0.918

Table 5: Correlation of human judgments against ROUGE-LSum, BLEU, BERTScore (and their extracted versions), and LLM-as-a-Judge (LLM-J).

Model	CW	ENT	FIB	IE	LOG	PE	QA	RC	ST	SUM	TC	TM	MT/TL	AVG	AVG*
GPT-4o	1.93	2.14	1.77	2.14	1.92	1.81	2.16	2.21	1.99	1.98	2.23	2.02	2.3	2.05	2.03
Gemini 2.0 Flash	1.96	2.00	1.55	2.15	1.91	2.18	2.20	2.27	1.85	1.99	2.03	1.98	2.24	2.02	2.01
Iron Horse GV V5a	1.90	2.14	1.35	2.17	1.88	2.56	2.12	2.05	1.82	1.89	1.90	2.02	2.51	2.02	1.98
DeepSeek V3	1.7	2.21	1.52	2.1	1.88	2.32	2.01	2.11	1.83	1.95	2.04	2.02	2.21	1.99	1.97
Claude Sonnet 3.5	1.85	2.07	1.32	2.08	1.8	2.42	2.09	2.18	1.88	1.95	1.79	2.09	2.37	1.99	1.96
Grok-2-latest	1.94	2.07	1.29	2.10	2.01	2.15	2.04	2.22	1.59	1.98	2.07	1.86	2.10	1.96	1.94
Nuha v2	1.86	1.86	1.39	1.99	1.84	2.37	1.91	2.20	1.59	1.95	2.20	1.86	1.96	1.92	1.92
Qwen-2.5 32b	1.85	1.93	1.39	1.88	1.82	1.88	1.79	2.02	1.57	1.96	1.77	1.78	1.74	1.8	1.8
Mistral-saba-latest	1.82	1.93	1.39	1.98	1.68	1.43	1.98	2.12	1.6	1.84	1.95	1.84	2.06	1.81	1.79
Gemma2 9B	1.78	2.29	1.26	1.94	1.67	1.61	1.72	2.15	1.41	1.96	1.72	1.62	1.67	1.75	1.76
c4ai-aya-expans-32b	1.71	1.93	1.03	1.90	1.58	2.01	1.8	1.99	1.20	2.02	1.64	1.87	2.14	1.75	1.72
Command R+	1.76	1.79	0.94	1.96	1.54	1.7	1.85	2.03	1.41	1.74	1.57	1.82	2.35	1.73	1.68
Amazon Nova Pro	1.77	2.07	1.13	1.81	1.54	1.35	1.81	1.81	1.49	1.65	1.68	1.95	2.18	1.71	1.67
Yehia-7B preview	1.79	2.14	0.9	1.89	1.46	1.34	1.73	2.06	1.17	1.83	1.62	1.83	2.02	1.68	1.65
Fanar-C-1-8.7B	1.70	1.93	0.90	1.88	1.53	1.96	1.72	1.79	0.95	1.86	1.52	1.71	-	-	1.62
Jais-family 13b-chat	1.80	1.86	0.52	1.62	1.39	2.42	1.49	1.85	0.66	2.05	1.11	1.57	-	-	1.53
SILMA-9B Instruct-v1.0	1.67	1.57	0.97	1.63	1.50	1.31	1.46	2.17	1.01	1.73	1.77	1.5	1.84	1.55	1.52
Command-r7b 12-2024	1.57	1.79	0.65	1.62	1.45	1.56	1.64	1.94	1.05	1.58	1.15	1.67	2	1.51	1.47
Mistral large	1.52	1.21	1.13	1.65	1.54	1.50	1.57	1.86	1.04	1.52	1.51	1.16	1.47	1.44	1.43
AceGPT-v2-8B-Chat	1.56	1.71	0.58	1.73	1.27	0.92	1.62	1.85	0.88	1.74	0.95	1.54	1.72	1.39	1.36
DBRX-instruct	0.73	0.93	0.33	1.10	0.81	0.96	1.09	1.4	0.85	0.78	1.18	0.67	1.14	0.92	0.90
Aragpt2-mega	-	0.00	0.03	-	0.05	-	0.12	0.25	0.02	0.15	0.15	0.29	-	-	-

Table 6: **LLM-as-a-judge evaluation.** “-” indicates that the model exceeded token limits and did not complete the category. List of categories: CW (Creative Writing), ENT (Entailment), FIB (Fill in the Blank), IE (Information Extraction), LOG (Logic), PE (Program Execution), QA (Question Answering), RC (Reading Comprehension), ST (Sequence Tagging), SUM (Summarization), TC (Text Classification), TM (Text Manipulation), MT/TL (Machine Translation/Transliteration), AVG (Average), AVG* (Average w/o Translation).

religious questions, ability to chat in a multi-turn scenario, propensity to hallucinate, tool usage, structured output generation, and many others. We plan to address many of these aspects in the next iteration of BALSAM with new test sets.

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A Dataset Statistics

Tables 7 and 8 present BALSAM Phase 1 and 2 benchmark dataset statistics, respectively.

No.	Category	Task	Test	Dev
1	Creative Writing	Definition Generation	22	22
		Dialogue Generation	146	65
		Explanation	64	21
		Instruction Generation	10	4
		Misc.	21	9
		News Article Generation	12	12
		Poem Generation	25	9
		Question Generation	1146	483
		Question Rewriting	48	20
		Sentence Composition	235	94
		Sentence Compression	21	10
		Story Composition	430	207
		Subject Generation	497	232
		Text Completion	119	46
		Text Generation	130	92
		Wrong Candidate Generation	233	93
2	Entailment	Textual Entailment	14	13
3	Fill in the Blank	Fill in The Blank	31	10
4	Information Extraction	Coreference Resolution	18	7
		Disease Mention Identification	10	9
		Keyword Extraction	47	43
		Named Entity Recognition	161	74
		Question Understanding	22	10
		Relation Extraction	10	9
5	Logic	Extracting Required Information	335	146
		Cause Effect Classification	39	18
		Coreference Resolution	13	6
		Misc.	69	29
		Predictive Analysis	10	10
		Riddle Solving	48	25
6	Translation/Transliteration	Sentence Ordering	18	8
		Dialect Translation	1200	600
		Machine Translation	1810	646
7	Program Execution	Transliteration	220	220
		Program Execution	646	268
8	Question Answering	Answering Given Question	2600	1484
		Question Decomposition	10	2
9	Reading Comprehension	Reading Comprehension	492	218
10	Sequence Tagging	Grammar Detection	277	129
		Keyword Extraction	58	20
11	Summarization	Text Summarization	618	399
		Answer Extraction	10	5
		Subject Generation	10	3
		Subject Identification	10	8
		Topic Identification	23	18
12	Text Classification	Command Interpretation	23	23
		Dialect Identification	27	27
		Emotion Detection	10	9
		Intent Classification	10	4
		Offensive Language Detection	21	11
		Problem Identification	10	8
		Sarcasm Detection	17	12
		Sentiment Analysis	10	2
		Text Categorization	56	23
13	Text Manipulation	Gender Rewriting	347	119
		Grammar Correction	269	202
		Intent Classification	18	5
		Paraphrasing	117	58
		Question Rewriting	100	34
		Text Simplification	98	41
		Total	13,121	6,434

Table 7: BALSAM Phase 1 benchmark dataset statistics

No.	Category	Task	Test	Dev
1	Creative Writing	Dialogue Generation	72	30
		Explanation	25	10
		Text Completion	50	20
		Text Continuation Evaluation	10	10
2	Entailment	Duplicate Question Identification	20	20
		Semantic Similarity	150	150
		Textual Entailment	305	150
3	Factuality	Answer Verification	50	20
		Answerability Classification	25	10
		Claim Verification	170	95
		Text Classification	100	49
3	Fill in the Blank	Fill in The Blank	25	10
		Discourse Connective Identification	10	4
4	Information Extraction	Disease Mention Identification	10	10
		Named Entity Recognition	10	10
		Entity Categorization	10	10
		Entity Recognition and Gender Identification	30	30
		Entity Relation Classification	25	10
		Extracting Required Information	35	20
		Text Classification	188	44
5	Logic	Cause Effect Classification	350	175
		Coherence Classification	50	20
		Commonsense Validation	130	80
		Evidence Evaluation	50	25
		Logical Reasoning	30	30
		Natural Language Inference	35	35
6	Translation/Transliteration	Machine Translation	12890	3225
7	Program Execution	Program Execution	25	10
8	Question Answering	Answering Given Question	4979	2117
9	Reading Comprehension	Answer Verification	25	10
		Answerability Classification	75	30
		Question Understanding	25	10
		Reading Comprehension	350	250
10	Sequence Tagging	Sequence Tagging	100	25
10	Text Classification	Dialect Identification	490	228
		Dialogue Act Recognition	25	10
		Emotion Detection	100	100
		Ethics Classification	50	20
		Hate Speech Detection	80	80
		Offensive Language Detection	200	110
		Query Classification	50	24
		Question Categorization	10	10
		Question Understanding	25	10
		Review Rating Prediction	30	30
		Sarcasm Detection	70	70
		Sentiment Analysis	605	509
		Text Categorization	235	110
		Text Classification	1584	983
		Topic Identification	10	10
13	Text Manipulation	Diacritization	300	250
Total			24298	9,308

Table 8: BALSAM Phase 2 benchmark dataset statistics.

B Examples of Samples

Figure 2 shows examples of some prompts and responses for the different categories.

Category/Dataset	Sample Input	Output
Creative Writing		
task568_circa_question_generation		في هذه المهمة ، ستعمل جوليا ، وهما هذان هي اقترح سؤال مقبول لتلك الإجابة. أنهيب للسيتيما كثيرا
Entailment		
enli_explain_the_choice		الترح لماذا يمكن وصف العلاقة بين الجنسين التاليين بأنها تناقض. الجملة 1: رجل عازي الصدر لديه وشتم وسروال أسود وحذاء أسود يقف على منصة أمام مراء. الجملة 2: الرجل حيا مستغرق على السطح
Fill in the Blank		
task1217_atomic_answer_generation		في هذه المهمة، ستلقي جملة تحتوي على كلمة مفقودة يمكن أن تكون جسماً أو شخصاً، و / أو فعلاً. قم بملء الفراغ بكلمة مناسبة. على الرغم من وجود وجود العديد من الإجابات الصحيحة لكل جملة، يجب عليك كتابة إجابة واحدة فقط. "بعض كرسى" سيرجل قريبا
Information Extraction		
task1479_organization_entity_extraction_bt c_corpus		في هذه المهمة، سوف تعطى جملة. من الموقع أن تعرف اسم الشركة أو المنظمة. على الرغم من أنه قد تكون هناك عدة أجوبة صحيحة ، يلزم أن تكتب واحدا منها. قطعة عظمية لكريستين ميلان في عطاراد اليوم. على الناس أن يدركوا أن الحرب الليبرالي يخطط للتسجيل على الموقع http://t.co/NRCH1cvs1
Logic		
ropes.plain_no_background		كانت مياه بحيرة الرب منخفضة هذا الربيع على الرغم من أن البحيرة عادة ما تكون على مستوى كامل في أواخر الربيع. كان الحراس يخططون لترك الماء خارج البحيرة حتى لا ينفج الجليد المراقبي وغيرها من الهياكل حول البحيرة. وكانت مستويات المياه مرتفعة بما فيه الكفاية في الخريف، فبدأت في خفض مستوى المياه بمقدار 10 أقدام، وانظروا إلى القفزة الواردة أعلاه، يرجى الإجابة على السؤال التالي على النحو الصحيح: متى تنخفض مستويات المياه في الربيع أو خلال الصيف؟ خلال الصيف
Program Execution		
task852_synthetic_multiply_odds		في هذه المهمة سوف تحصل على قائمة من القوائم، تحتوي على أرقام صحيحة. لكل قائمة داخلية في قائمة المدخلات، يجب أن تقرب كل الأرقام القريبة في تلك القائمة. ينبغي أن يكون الناتج قائمة بالأعداد الصحيحة بنفس طول عدد القوائم في قائمة المدخلات. إذا لم يكن هناك أرقام مفردة في قائمة داخلية يجب أن تخرج 0 لتلك القائمة. [[36, -23, -41], [27, 46, 37, 31, -24], [5, 50, -1, 38, 2, -3]]
Question Answering		
arcd.qa		خضعت إسبانيا بسبب موقعها لمؤثرات خارجية كثيرة منذ عصور ما قبل التاريخ وحتى بروجيها كدولة. خرجت إسبانيا كدولة موحد في القرن الخامس عشر وذلك بعد توحيد الممالك الكاثوليكية والسيطرة على كامل شبه الجزيرة الأيبيرية في 1492. من ناحية أخرى، كانت إسبانيا مصدر نفوذ هام في مناطق أخرى خلال العصور الحديثة، عندما أصبحت إمبراطورية عالمية خلفت إرثا يضم أكثر من 500 مليون ناطق بالإسبانية، مما يجعلها ثاني اللغات الأكثر استخداما. وفقا للنص السابق، متى خرجت إسبانيا كدولة موحدة؟ في القرن الخامس عشر
Reading Comprehension		
xp3_mliqa_mliqarar_test_cqreferringa		
Sequence Tagging		
spelling_error_dataset_card		
Summarization		
hellaswagTopic_of_the_context		
Text Classification		
aeineima_Saudilrony_editedCNIsirony1		
Text Manipulation		
task927_yelp_negative_to_positive_style_transfer		
Translation-Transliteration		
arbnl_Arabizi_Transliteration.translation		

Figure 2: Samples from different categories.

C Examples of Prompts

Figure 3 shows some prompt templates that we used to create some of the datasets.

حدد إن كانت التغريدة الآتية: “{{text}}” عادية أم مزعجة {{answer}} هذه التغريدة
Translation: Specify if the following tweet: “{{text}}” is normal or spam The tweet is {{answer}}
أريد تصنيف التغريدة الآتية: “{{text}}” لمعرفة إذا كانت عادية أم مزعجة {{answer}} التغريدة التي ذكرتها
Translation: I want to classify the following tweet: “{{text}}” to know if it is normal or spam The tweet you provided is {{answer}}
“{{text}}” بالنسبة للرسالة السابقة، هل هي عادية أم دعائية {{answer}} الرسالة
Translation: “{{text}}” concerning the preceding message, is it normal or an advert The message is {{answer}}
وصلتني الرسالة الآتية: “{{text}}” يا ترى هل هي "عادية" أم "غير مرغوب فيها" {{answer}} أظن أن الرسالة
Translation: I received the following message: “{{text}}” I wonder if it is normal or unsolicited The think the message is {{answer}}

Figure 3: Example prompts for the Arabic tweet classification task.

D Models

D.1 Open-source Models

- **AceGPT-v2-8B-Chat (Huang et al., 2024)**: A fine-tuned Arabic dialogue model based on LLaMA2, designed for chat-style interactions in Arabic.
- **Aragpt2-mega (1.5B) (Antoun et al., 2021)**: A large-scale Arabic GPT-2 model designed for generating and understanding Arabic text.
- **c4ai-aya-expanse-32b (Dang et al., 2024)**: A multilingual large language model supporting 23 languages, including Arabic, with strong performance across diverse tasks.
- **Command R+ (104B):⁵** A multilingual model optimized for retrieval-augmented generation (RAG), reasoning, and task completion, with general Arabic support.
- **Command-r7b 12-2024:⁶** A compact and efficient version of the Command family of models, designed for general-purpose instruction following and language generation.
- **DeepSeek V3 (685B) (DeepSeek-AI, 2024)**: A multilingual Mixture-of-Experts model for reasoning, coding, and language understanding.
- **Gemma2 9B (Gemma Team et al., 2024)**: A multilingual language model from Google.
- **Jais-family 13b-chat (Sengupta et al., 2023)**: A bilingual Arabic-English model trained on 395B tokens, optimized for long-sequence handling.

⁵<https://huggingface.co/CohereLabs/c4ai-command-r-plus>

⁶<https://huggingface.co/CohereLabs/c4ai-command-r7b-12-2024>

- **qwen-2.5 32b (Yang et al., 2024):** A high-capacity language model with strong performance in Chinese and English and expanding capabilities in other languages, including Arabic.
- **SILMA-9B Instruct-v1.0:**⁷ A 9-billion-parameter Arabic language model built on Google’s Gemma architecture, fine-tuned for instruction-following tasks.
- **Yehia-7B preview:**⁸ A bilingual model designed for Arabic and English, capable of instruction-following and engaging in natural dialogue.
- **Fanar (Fanar Team et al., 2025):** It comes with two 7B and 9B parameter LLMs trained on nearly 1 trillion tokens. The models are designed to support both Arabic and English.
- **Mistral large :**⁹ A multilingual model with 123B parameters by Mistral AI.
- **DBRX-instruct (132B):**¹⁰ An instruction-tuned transformer developed by Databricks for high-quality reasoning and generation.

D.2 Closed-Source Models

- **Nuha v2 :**¹¹ Nuha is an advanced, culture-aware AI assistant infused with pre-training and fine-tuning to understand Arabic nuances. With Nuha that is 40B parameter .
- **Iron Horse Gamma Velorum V5a:**¹² A closed-source MoE model with 1.1T - 2.3T parameters based on the request. It supports more than 25 language. .
- **Amazon Nova Pro (et al., 2025)** A multilingual model by Amazon Bedrock designed for commercial applications.
- **Mistral-saba-latest (24B):**¹³ An Arabic fine-tuned variant of the Mistral model.
- **Grok-2-latest (314B MoE):** A closed-source model by xAI, designed for reasoning and factual recall.
- **Claude Sonnet 3.5:**¹⁴ A multilingual and instruction-capable model by Anthropic, estimated at over 130B parameters.
- **Gemini 2.0 Flash:** A lightweight variant of Gemini 2.0 optimized for speed and extended context.
- **GPT-4o (OpenAI et al., 2024):** OpenAI’s model supporting multimodal and multilingual input, including Arabic.

⁷<https://huggingface.co/silma-ai/SILMA-9B-Instruct-v1.0>

⁸<https://huggingface.co/Navid-AI/Yehia-7B-preview>

⁹<https://huggingface.co/mistralai/Mistral-Large-Instruct-2407>

¹⁰<https://huggingface.co/databricks/dbrx-instruct>

¹¹<https://nuha.ai/>

¹²<https://www.ironhorse.ai/>

¹³<https://mistral.ai/news/mistral-saba>

¹⁴<https://www.anthropic.com/news/claude-3-5-sonnet>

E Example LLM Outputs with the Same Meaning

Consider the following Question Answering example where the correct answer is باريس (Paris). While SILMA 9B just answered with باريس only, the other model responses were much more verbose. Consider the answer of c4ai-aya-expanse-32b:

الجواب هو: باريس. وفقًا للفقرة، بدأت مرحلة تتابع الشعلة في فرنسا من باريس، حيث انعقدت في ٧ أبريل، وبدأت على المستوى الأول من برج إيفل، وانتهت في ملعب تشارلتي.

– **Translation:** The answer is: Paris. According to the piece, the journey of the torch started in France from Paris on April 7 where it started from the first level of the Eiffel Tower and ended at the Charléty stadium.

F Prompts for LLM-Based Evaluation

Here is the prompt we used to extract the correct answer only from the LLM output:

```
"""Given the following prompt:

{prompt}

And the following automatically generated output:

{response}

Extract the answer from the automatically generated output ONLY WITHOUT any modification. Remove all non-related text from the answer. Do not put any additional text. If there are multiple answers, extract the first one only.

"""
```

Listing 1: Prompt for LLM-based answer extraction.

Here is the prompt that we used for LLM as a judge:

```
You are an impartial and expert judge evaluating the quality of text generated by another AI model. Your task is to score the generated output based on the original prompt and a provided ground truth answer, following a specific scoring rubric.

You will be provided with three pieces of information:
1. The original prompt given to the generative model.
2. The ground truth answer, representing the ideal or expected output.
3. The actual output generated by the generative model.

Evaluate the generated output by comparing it to the ground truth, considering how well it addresses the original prompt.

Scoring Rubric:
* Score 0: The automatically generated output is completely wrong, irrelevant, or unrelated to the prompt and ground truth.
* Score 1: Poor answer. The output attempts to address the prompt but contains significant errors, is largely incomplete, or is difficult to understand. It shows little resemblance to the ground truth.
* Score 2: Acceptable but different. The output is somewhat correct or addresses parts of the prompt reasonably well, but it differs significantly from the ground truth. It might be missing details present in the ground truth, include extra information not in the ground truth, or present the information in a substantially different structure or style, but it is still a valid (though not ideal) response to the prompt.
* Score 3: Perfect or almost perfect. The output is accurate, complete, and closely matches the ground truth in content and style, effectively answering the original prompt. Minor differences in wording or formatting that do not affect the meaning or quality are acceptable for a score of 3.

Output Format:
Your output must be *only* a JSON object containing two keys:
1. `score`: An integer between 0 and 3 based on the rubric above.
2. `explanation`: A brief, concise string explaining *why* you assigned that score, referencing the differences or similarities between the generated output and the ground truth in the context of the prompt.

Example Output JSON:
{
  "score": 3,
```

```

"explanation": "The generated output is accurate and complete, closely matching the ground truth."
}

[PROMPT]
{prompt}
[/PROMPT]

[GROUND TRUTH]
{reference answer}
[/GROUND TRUTH]

[GENERATED OUTPUT]
{response}
[/GENERATED OUTPUT]

```

Listing 2: LLM-as-a-Judge prompt.

G Human Evaluation Annotation Instructions

- هل الإجابة صحيحة عند مقارنتها مع الإجابة الأصلية (٠-٣)؟
- ٠: إجابة خاطئة تماما (لا تتطابق مع الإجابة الأصلية بأي شكل).
 - ١: إجابة خاطئة جزئيا (تحتوي على بعض العناصر الصحيحة ولكن بها أخطاء جوهرية).
 - ٢: إجابة صحيحة جزئيا (تعكس بعض المعنى الصحيح ولكنها تفتقر إلى الدقة أو التفاصيل المهمة).
 - ٣: إجابة صحيحة تماما (متطابقة أو مكافئة للإجابة الأصلية دون أي أخطاء)

Translation of instructions:

Is the answer correct when compared to the original answer (0-3)?

- 0: Completely wrong (does not match the original answer in any way).
- 1: Partially wrong (contains some correct elements but has significant errors).
- 2: Partially correct (conveys some correct meaning but lacks accuracy or important details).
- 3: Completely correct (identical or equivalent to the original answer with no errors).